

Testing of Hypothesis

F-test for equality of two population Variance!

If we want to test

- (i) Whether two independent samples x_i ($i=1, 2, 3, \dots, n_1$) and y_j ($j=1, 2, 3, \dots, n_2$) have been drawn from the normal population with the same variance σ^2 (say) or
- (ii) Whether the two independent estimates of the population variance are homogeneous or not.

under the null hypothesis (H_0) that
(i) $\sigma_x^2 = \sigma_y^2 = \sigma^2$; i.e. the population variances are equal.

or
(ii) Two independent estimates of the population variances are homogeneous.

The Statistic F is given by:

$$F = \frac{S_x^2}{S_y^2} \rightarrow \textcircled{1}$$

Where $S_x^2 = \frac{1}{(n_1-1)} \sum_{i=1}^{n_1} (x_i - \bar{x})^2$ and $S_y^2 = \frac{1}{(n_2-1)} \sum_{j=1}^{n_2} (y_j - \bar{y})^2$
 $\rightarrow \textcircled{2}$

Where S_x^2 and S_y^2 are unbiased estimates of the common population variance σ^2 obtained from two independent samples and it follows F-distribution with (v_1, v_2) d.f. where $v_1 = n_1 - 1$; $v_2 = n_2 - 1$.

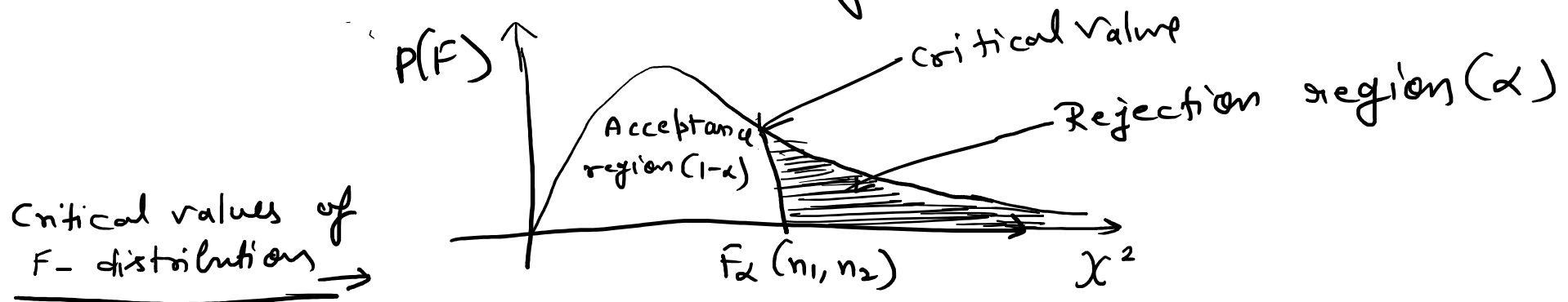
Remarks ① In (i); greater of the two variances S_x^2 and S_y^2 is to be taken in the numerator and n_1 corresponds to the greater variance.

By comparing the calculated value of F obtained by (i) for the two given samples, with the tabulated value of F for (n_1, n_2) d.f. at certain level of significance (5% or 1%). H_0 is either rejected or accepted.

(2) Critical values of F-distribution - The available F-table give the critical values of F for the right tailed test. i.e. the critical region is determined by the right tail areas. Thus the significant value $F_{\alpha}(n_1, n_2)$ at level of significance α and (n_1, n_2) d.f. is determined by

$$P[F > F_{\alpha}(n_1, n_2)] = \alpha$$

as shown in the diagram:-



③ We have the following reciprocal relation between the upper and lower α -significant points of F-distribution:-

$$F_{\alpha}(n_1, n_2) = \frac{1}{F_{1-\alpha}(n_2, n_1)}$$

$$\Rightarrow F_{\alpha}(n_1, n_2) \times F_{1-\alpha}(n_2, n_1) = 1$$

④ The critical values of F

① for left tail test $H_0: \sigma_1^2 = \sigma_2^2$
against $H_1: \sigma_1^2 < \sigma_2^2$ are given by

$$F < F_{n-1, n_2-1}(1-\alpha)$$

(b) For two tailed test : $H_0 : \sigma_1^2 = \sigma_2^2$
against $H_1 : \sigma_1^2 \neq \sigma_2^2$ are given by

$$F > F_{n_1-1, n_2-1}(\alpha/2) \text{ and}$$

$$F > F_{n_1-1, n_2-1}(1-\frac{\alpha}{2})$$

Example: Two random samples drawn from normal populations are:

Sample-I	Sample-II
20, 16, 26, 27, 23, 22, 18, 24, 25, 19	27, 33, 42, 35, 32, 34, 38, 28, 41, 43, 30, 37

Obtain estimates of the variances of the two populations and test whether two populations have the same variance.

$$\bar{X} = \frac{\sum X}{n_1} = \frac{220}{10} = 22; \quad \bar{Y} = \frac{\sum Y}{n_2} = \frac{420}{12} = 35$$

$$S_1^2 = \frac{\sum (X - \bar{X})^2}{n_1 - 1} = \frac{120}{9} = 13.33$$

$$S_2^2 = \frac{\sum (Y - \bar{Y})^2}{n_2 - 1} = \frac{314}{11} = 28.545$$

Using F-test, we have

$$F = \frac{S_2^2}{S_1^2} = \frac{28.545}{13.33} \text{ where } S_2^2 > S_1^2$$

For $v_1 = 11$ and $v_2 = 9$, $F_{.05} = 3.11$

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Q. In one sample of 8 observations, the sum of the squares of deviations of the sample values from the sample mean was 84.4 and in other sample of 10 observations it was 102.6. Test whether this difference is significant at 5 per cent level, given that the 5 per cent point of F for $n_1 = 7$ and $n_2 = 9$ degrees of freedom is 3.29.

Solⁿ Here $n_1 = 8$, $n_2 = 10$, $\sum (x - \bar{x})^2 = 84.4$, $\sum (y - \bar{y})^2 = 102.6$

$$\therefore S_x^2 = \frac{1}{n_1 - 1} \sum (x - \bar{x})^2 = \frac{84.4}{7} = 12.057$$

$$S_y^2 = \frac{1}{n_2 - 1} \sum (y - \bar{y})^2 = \frac{102.6}{9} = 11.4$$

$[S_x^2 > S_y^2]$

Under $H_0: \sigma_x^2 = \sigma_y^2 = \sigma^2$

i.e. the estimates of σ^2 given by the samples are homogeneous

Test Statistic :- $F = \frac{S_x^2}{S_y^2} = \frac{12.057}{11.4} = 1.057$

Tabulated $F_{0.05}$ for (7, 9) d.f. is 3.29

Since calculated $F = F_{cal.} < F_{0.05}$

H_0 may be accepted at 5% level of significance.

Q. Two random samples gave the following results:-

Sample	Size	sample mean	Sum of squares of deviations from mean
1	10	15	90
2	12	14	108

Test whether the sample come from the same normal population at 5% level of significance.

[Given: $F_{0.05}(9,11) = 2.90$, $F_{0.05}(11,9) = 3.10$ (approx.) and $t_{0.05}(20) = 2.086$, $t_{0.05}(22) = 2.07$.]

Solⁿ A normal population has two parameters; mean μ and variance σ^2 .
To test if two independent samples have been drawn from

from the same normal population! -

We have to test

(i) The equality of population mean.

(ii) The equality of population variance.

Null Hypothesis: The two samples have been drawn from the same normal population, i.e. $H_0: \mu_1 = \mu_2$ and $\sigma_1^2 = \sigma_2^2$

* Equality of means will be tested by applying t-test

* Equality of variances will be tested by applying F-test

Since t-test assumes $\sigma_1^2 = \sigma_2^2$;

we shall first apply F-test and then t-test.

We are given

$$n_1 = 10, \quad \bar{x}_1 = 15, \quad \sum (x_1 - \bar{x}_1)^2 = 90$$

$$n_2 = 12, \quad \bar{x}_2 = 14, \quad \sum (x_2 - \bar{x}_2)^2 = 108.$$

F-test:- $S_1^2 = \frac{1}{n_1 - 1} \sum (x_1 - \bar{x}_1)^2 = \frac{90}{9} = 10$

$$S_2^2 = \frac{1}{n_2 - 1} \sum (x_2 - \bar{x}_2)^2 = \frac{108}{11} = 9.82$$

Since $S_1^2 > S_2^2$; under $H_0: \sigma_1^2 = \sigma_2^2$; the
test statistic is

$$F = \frac{S_1^2}{S_2^2} \sim F(n_1-1, n_2-1) = F(9, 11)$$

$$\therefore F = \frac{S_1^2}{S_2^2} = \frac{10}{9.82} = 1.018$$

Tabulated $F_{0.05}(9, 11) = 2.90$

$$\Rightarrow \text{Cal. } F < \text{Tab. } F$$

$\Rightarrow$ it is not significant.

Hence null Hypothesis H_0 of equality of population variance may be accepted. $\Rightarrow (\sigma_1^2 = \sigma_2^2)$

Since $\sigma_1^2 = \sigma_2^2$

We can now apply t-test for testing $H_0: \mu_1 = \mu_2$

t-test: Under $H_0: \mu_1 = \mu_2$

against alternative hypothesis $H_1: \mu_1 \neq \mu_2$

test statistic :- $t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{S^2(\frac{1}{n_1} + \frac{1}{n_2})}} \sim t_{n_1+n_2-2} = t_{20}$

$$\text{where } S^2 = \frac{1}{n_1 + n_2 - 2} \left[\sum (x_1 - \bar{x}_1)^2 + \sum (x_2 - \bar{x}_2)^2 \right] = \frac{1}{20} (90 + 108) = 9.9$$

$$t = \frac{15 - 14}{\sqrt{9.9(\frac{1}{10} + \frac{1}{12})}} = 0.742.$$

Tabulated $t_{0.005}$ for 20 d.f.
 $= 2.086$

$$\Rightarrow |t| < t_{0.05}$$

$\Rightarrow$ Not significant

$\Rightarrow H_0: \mu_1 = \mu_2$ may be accepted

$\Rightarrow$ Since both hypotheses $H_0: \mu_1 = \mu_2$
and $H_0: \sigma_1^2 = \sigma_2^2$ are accepted

$\Rightarrow$ The given samples have been drawn from the
same normal population.